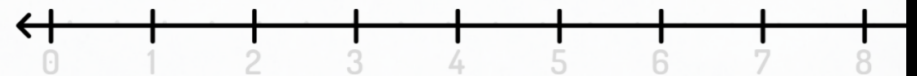


Numerical Optimization

BS./BSc.

in Applied AI and Data Science



IIT Jodhpur | Sem 2 | 2025

Overview



- Review
- Convexity
- Convex set and convex function
- First-order and second-order characterization of convexity
- Gradient descent algorithm



Review of Lecture 3

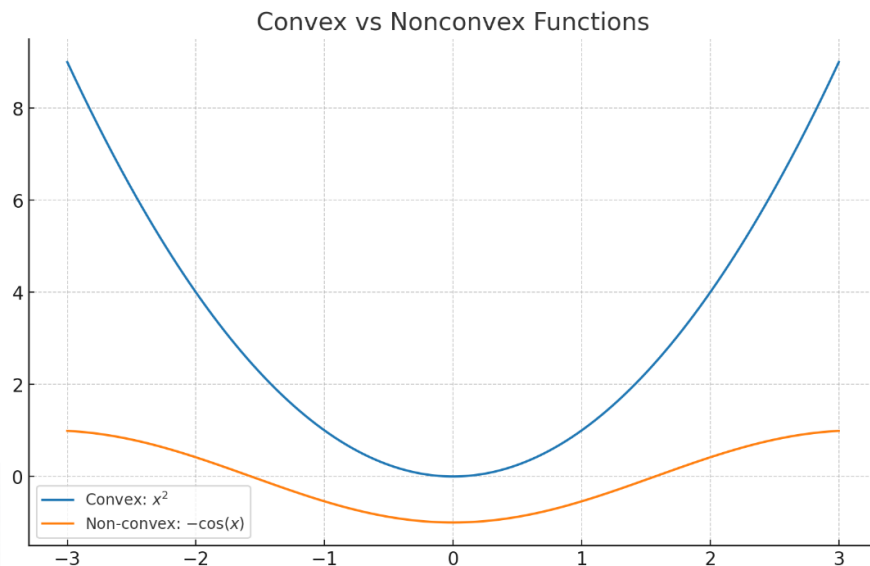


- First order necessary condition for local optimality: Fermat's rule
- Second order Hessian condition
- Positive definiteness of matrices

Why Convexity?



- In general optimization, local minima may not be global.
- First order condition may not be sufficient
- For convex problems: every local minimum is global and first order condition are sufficient.
- Many ML problems (linear regression, logistic regression, SVMs) are convex.



Convex Sets

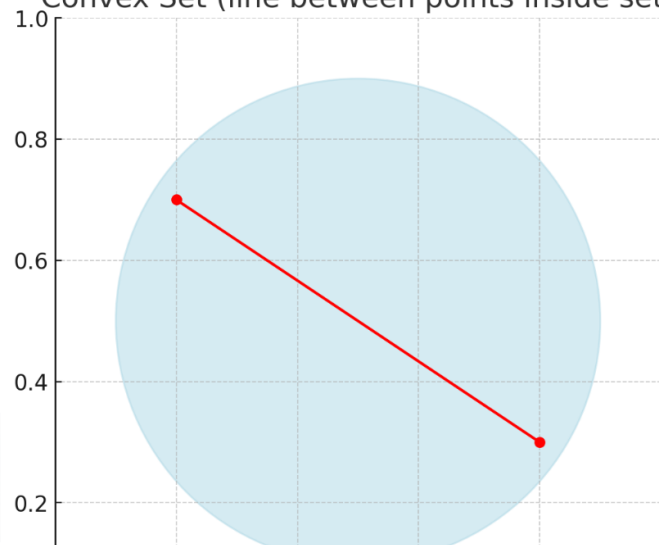


Definition

A set $C \subseteq \mathbb{R}^n$ is convex if for all $x, y \in C$, and $\theta \in [0, 1]$:

$$\theta x + (1 - \theta)y \in C.$$

Convex Set (line between points inside set)



Convex Functions

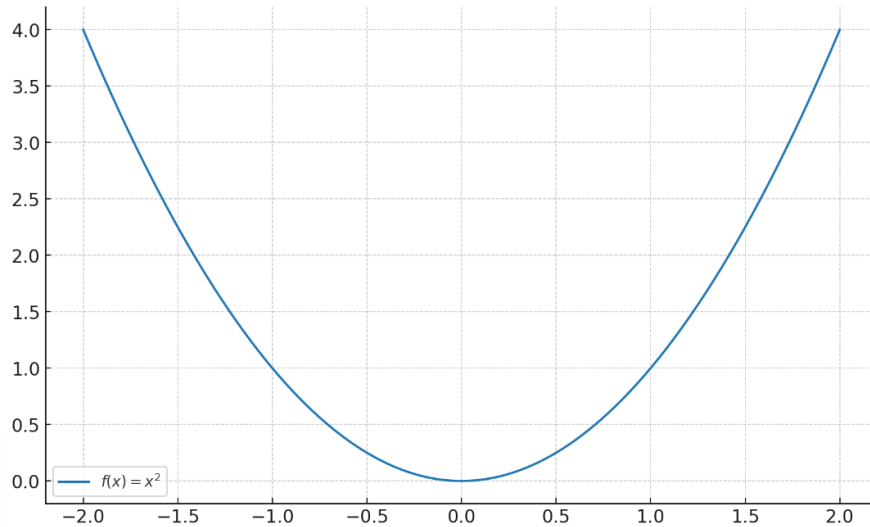


Definition

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if for all x, y and $\theta \in [0, 1]$:

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y).$$

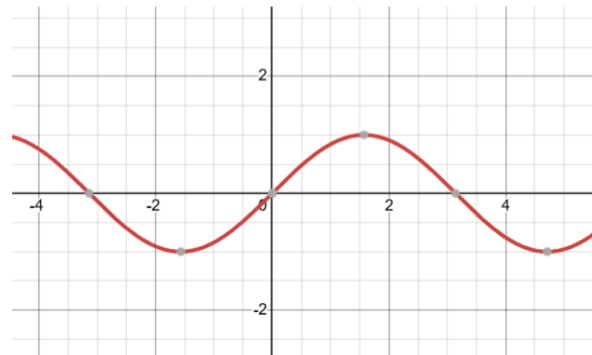
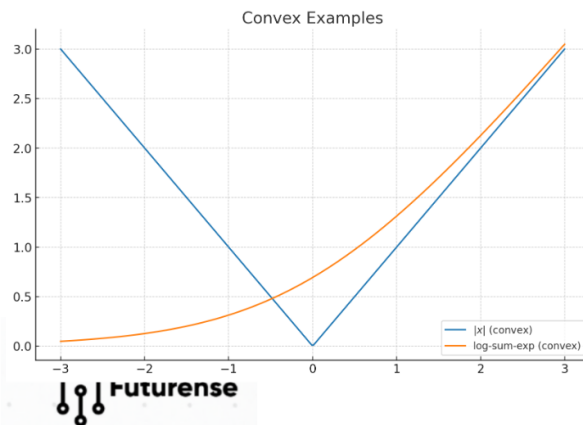
Convex Function: Line above curve



Examples



- Convex: $f(x) = x^2$, $f(x) = \|x\|_2$, $f(x) = e^x$, $f(x) = \log(\sum_{i=1}^n e^{x_i})$.
- Non-convex: $f(x) = -x^2$, $f(x) = \sin(x)$.
- $f : \mathbb{R} \rightarrow \mathbb{R}$, defined by $f(x) = ax^2 + bx + c$ where $a > 0$.
- In general, for a symmetric positive semidefinite matrix $A \in \mathbb{R}^{n \times n}$, the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, defined by $f(x) = x^T A x + b^T x + c$ where $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$ is a convex function.



First-Order Characterization

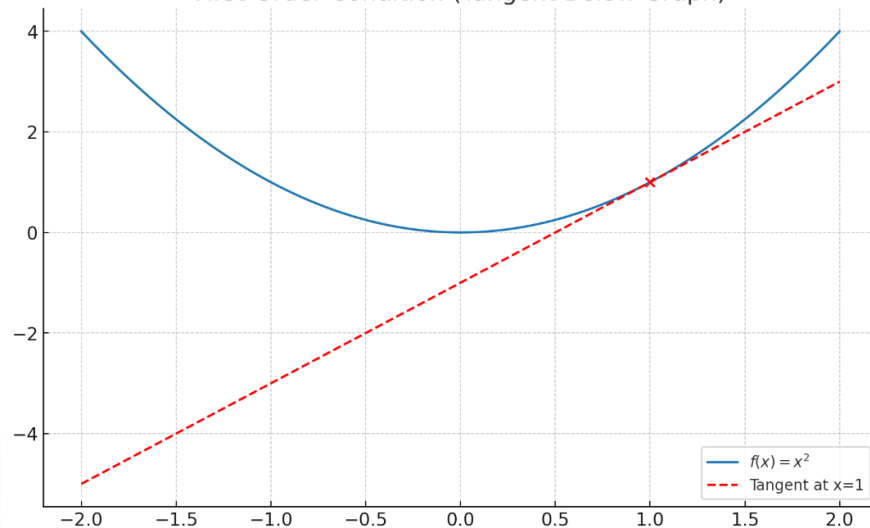


Theorem

A differentiable function f is convex iff

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y.$$

First-Order Condition (Tangent Below Graph)



Important implications of convexity



- Local solutions are global
- First order conditions are sufficient



Second-Order Condition



- If f is twice differentiable, then f is convex iff

$$\nabla^2 f(x) \succeq 0 \quad \forall x.$$

- A quadratic function is a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ which can be represented as

$$f(x) = \frac{1}{2}x^T A x + b^T x + c,$$

where $A \in \mathbb{R}^{n \times n}$, is a symmetric matrix $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$.

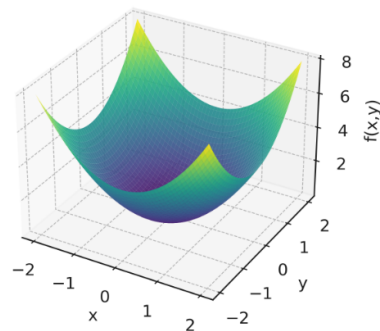
- For this function $\nabla f(x) = Ax + b$ and $\nabla^2 f(x) = A$.
- Thus a quadratic function is convex if and only if the defining matrix A is symmetric positive semidefinite.



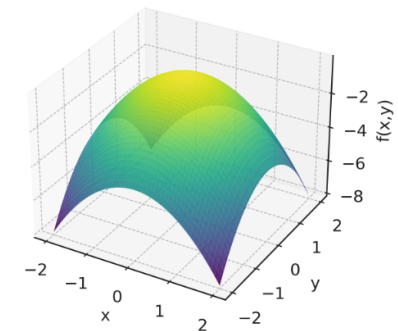
Hessian Shapes



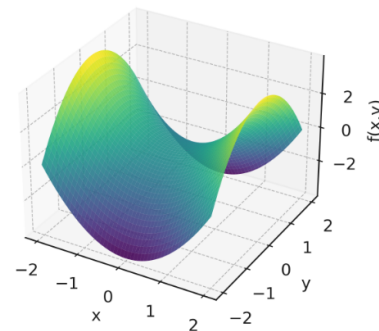
Positive Definite Hessian (Bowl)



Negative Definite Hessian (Cap)



Indefinite Hessian (Saddle)



Positive definite and semidefinite matrices and some examples



- Positive definite does not mean positive entries.
- How to know if a matrix is positive definite or semidefinite?
 - If all eigenvalues are nonnegative(positive)-It is an if and only if criteria.
 - If a matrix is positive semidefinite (definite), then trace and determinant is nonnegative (positive). For a 2×2 , matrix this is an if and only if criteria.
 - Positive definite if and only if all principle minors are positive.
 - Diagonally dominant matrix.
- Consider the function $f(x, y) = 8x + 12y + x^2 - 2y^2$
- Consider the function $f(x, y) = 100(y - x^2)^2 + (1 - x)^2$

Role of Convexity in Optimization



- Guarantees global optimality.
- Convex problems can be solved efficiently.
- Subgradient methods, duality, and many algorithms rely on convexity.





- Least squares regression: convex quadratic objective. Given $X \in \mathbb{R}^{m \times n}$, $y \in \mathbb{R}^m$:

$$f(w) = \|Xw - y\|_2^2 = (Xw - y)^T (Xw - y)$$

- Logistic regression: convex log-loss. For data $\{(x_i, y_i)\}_{i=1}^n$, $y_i \in \{-1, +1\}$:

$$f(w) = \sum_{i=1}^n \log(1 + e^{-y_i w^T x_i})$$

- Support Vector Machines (SVMs): convex hinge loss. For data $\{(x_i, y_i)\}_{i=1}^n$, $y_i \in \{-1, +1\}$:

$$L(w) = \sum_{i=1}^n \max(0, 1 - y_i w^T x_i)$$

Mini Exercises



- 1 Show $f(x) = \max(0, 1 - x)$ is convex.
- 2 Prove/disprove: $f(x) = x^3$ is convex.
- 3 Show least squares loss is convex in w .
- 4 Challenge: Show logistic regression loss is convex.



Why Numerical scheme?



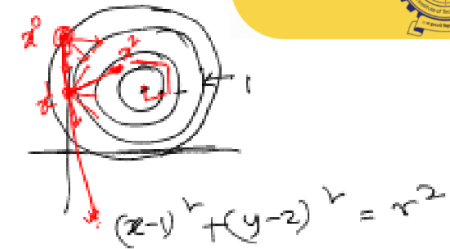
- Our main aim is to find critical points, that is, points that satisfy $\nabla f(x) = 0$. In general, it is difficult to solve them analytically when the equations are nonlinear.
- Many optimization problems, including convex optimization problems, have no closed-form solution.
- For example, consider $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = (x - 1)e^x - x$. We have $f'(x) = 0 \Rightarrow xe^x = 1$, which cannot be solved analytically.
- We thus use iterative methods to compute the critical points.
- Our aim is to generate a sequence $\{x_n\}_n$ such that $x_n \rightarrow x^*$ in certain sense with $\nabla f(x^*) = 0$.
- Algorithms vary in the way one chooses to move from one iteration to the next.



Line search



- 1: Initialize $x^0 = (x_1^0, x_2^0, \dots, x_n^0)$, tolerance tol , $k = 0$.
- 2: **while** $\|\nabla f(x^k)\| > tol$ **do**
- 3: **Direction:** Choose d^k such that $\nabla f(x^k)^T d^k < 0$.
- 4: **Step-size:** Choose α^k such that $f(x^k + \alpha^k d^k) < f(x^k)$ and satisfies certain condition.
- 5: **Update:** $x^{k+1} = x^k + \alpha^k d^k$
- 6: $k = k + 1$.
- 7: **end while**
- 8: Output x^k as a critical point of $f(\cdot)$.



- Gradient descent method / steepest descent method is the line search method where $d_k = -\nabla f(x_k)$ for all iterations k .

Gradient descent method



- 1: Initialize $x^0 = (x_1^0, x_2^0, \dots, x_n^0)$, tolerance tol , $k = 0$.
- 2: **while** $\|\nabla f(x^k)\| > tol$ **do**
- 3: **Direction:** Choose $d^k = -\nabla f(x^k)$.
- 4: **Step-size:** Choose α^k such that $f(x^k + \alpha^k d^k) < f(x^k)$ using any line search algorithm.
- 5: **Update:** $x^{k+1} = x^k + \alpha^k d^k$
- 6: $k = k + 1$.
- 7: **end while**
- 8: Output x^k as a critical point of $f(\cdot)$.

Why gradient?



Theorem

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuously differentiable function. Let $x \in \mathbb{R}^n$ be such that $\nabla f(x) \neq 0$. Then

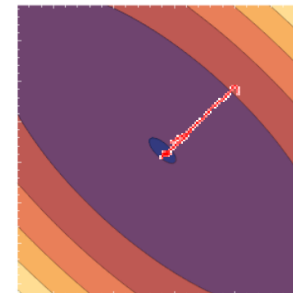
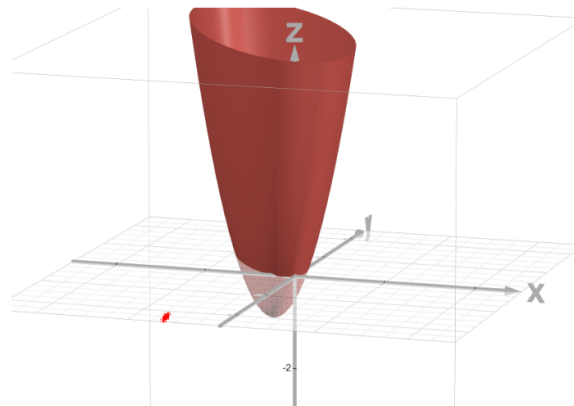
$$\langle \nabla f(x), -\frac{\nabla f(x)}{\|\nabla f(x)\|} \rangle \leq \langle \nabla f(x), d \rangle \quad \forall d \in \mathbb{R}^n \text{ with } \|d\| = 1$$

Steepest descent

Example



- Consider $f(x_1, x_2) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2$



- $\nabla f(x_1, x_2) = \begin{pmatrix} 1 + 2x_2 + 4x_1 \\ -1 + 2x_1 + 2x_2 \end{pmatrix}$, $\nabla^2 f(x_1, x_2) = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}$, $x^* = \begin{pmatrix} -1 \\ 1.5 \end{pmatrix}$

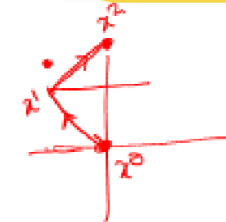
Summary



□ Amir Beck
□ Bazarra Shetty Gendhi
□ Numerical Optimization — Nocedal & Wright.

- Understood convexity: convex set and convex function
- First order and second order characterization of convexity
- Implication: Like solution are global, first order condition are sufficient
- Positive semidefinite and positive definite matrix
- Numerical scheme: iterative optimization for differentiable functions.
- Gradient descent

Example: Gradient descent with constant step size



- Let us consider $f(x_1, x_2) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2$
- $\nabla f(x_1, x_2) = \begin{pmatrix} 1 + 2x_2 + 4x_1 \\ -1 + 2x_1 + 2x_2 \end{pmatrix}$
- Let $x^0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $d^0 = -\nabla f(x^0) = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.
- Suppose we want to choose constant step size $\bar{\alpha} = 1$.
- $x^1 = x^0 + d^0 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$, $d^1 = -\nabla f(x^1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- $x^2 = x^1 + d^1 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$, and we continue...



- Doing manual calculation is tedious.
- Automate this for computer (refer to Google colab)

